OVERVIEW

CS2323: Computer Architecture

July 2025 – Nov 2025

Rajesh Kedia, CSE IITH, 2025

Discussion Topics

- Pre-requisites
- Course outline
- Evaluation
- Administrivia
- Suggested books

Pre-requisites

- Introduction to programming (ID1063/ID1303)

Course Outline

- Introduction to computer architecture, overall system components, OS and compilers, multiprocessors, trends, design objectives, etc.
- Instruction set architecture, assembly program, translation from C to assembly
- Processor design – Basic processor components, instruction encoding, building a processor, improving performance through pipelining and other techniques
- Caches and memory systems – cache hierarchy, various cache configurations, etc.
- Interfacing I/O devices
- Computer arithmetic – addition, multiplication, division, etc.
- Emerging research trends

Lab Exercises

- Assembly language coding in a simulator:
  - Understanding various instructions
  - Assembly code development for common mathematical functions
  - Implementing on a real hardware board

- Project connecting different concepts together:
  - Enhancing the simulator for pipelining (C/C++)
  - Or, Hardware implementation of processor (Verilog/VHDL)

Evaluation (tentative and can change)

- In-class short quizzes: 10-15.
  - ~5 quizzes to be conducted on Thursdays from 1pm.
- Homeworks: 5-10
- Lab assignments: 10-20
- Project: 15-25
- Mid-sem Exam: 20-30
- Final Exam: 25-40
- Lab quiz or Lab exam: 10-15
- Class participation: 2-7
- **Total**: 100

Administrivia

- Course credits: **4 (four) – the course will be lab-intensive**
- Instructor: Rajesh Kedia (+ invited lectures as appropriate)
- Course webpage: Check link on https://people.iith.ac.in/rkedia
- Lecture schedule: 3 hours a week, E slot (Tue, Thu, Fri), A-LH2
- Instructor contact hours: Just after the lectures or by appointment
- Lab venue, Schedule: To be announced
- Learn yourself and help others: discussion forum
- Only selected slides will be shared. Encouraged to read textbook
- No extensions to deadlines
- Moodle: Self-marking attendance, quizzes, assignment submission, discussion forum, etc.

Administrivia

- Attendance will be recorded for every class.
  - There are no direct marks for attendance. But, attending classes might help you do better in exams or quizzes and hence there are indirect benefits.
  - Any attendance proxy will also be considered as academic misconduct.
- Academic honesty MUST be followed: https://cse.iith.ac.in/academics/plagiarism-policy.html

References

- Computer Organization and Design, The Hardware/Software Interface (RISC-V edition) by David A. Patterson and John L. Hennessy
- The RISC-V Instruction Set Manual, Volume I: Unprivileged ISA, Document Version 20191213
- Digital Design and Computer Architecture (RISC-V edition), By David Harris and Sarah Harris
- Basic Computer Architecture (also available as Computer organization and architecture), version 2.1, By Smruti Ranjan Sarangi